

Music-Gen — Cycles 13-15

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Music-Gen — Cycles 13-15

Abstract

Cycles 13-15 covered the fork `ed041ef4c1dc` fanout with three parallel clones and its integration. Clone 0 grew the M-RULES-1 rules ledger from 28 to 76 rows by running the cycle-9 extractors over the two M-INGEST-1/breadth-second-seeds merged MusicXMLs, preserved the cycle-9 anchor row prefix byte-identically, and resolved the specific (salt 1, salt 4) arrangement collision the cycle-11 audit had named as the mechanical unlock for larger M-GEN-1 batches; the overall salt-collision pair count dropped from 5 to 4, with the residual salt-4 over-representation surfaced honestly as a research finding rather than tuned away. Clone 1 tightened the fanout concat seam so every merged row now validates against the cycle-10 SSoT `_ledger_schema.validate_event` and every candidate stream enforces per-milestone `ts` monotonic ordering with a `content_hash_tiebreak` on collision, closing the last drift surface on the campaign's ledger-write side; the concat surface, the emit surface, and the check surface now all consume the same schema module with `is-identity` of `REQUIRED_EVENT_FIELDS` verified live. Clone 2 attempted the cycle-3 M-DAW-SPIKE-1 documented-GAP fallbacks: GAP-1 (Ardour Lua MIDI-file import) redefined-GAP via `fluidsynth` pre-render plus hand-authored Ardour audio-region XML (`env_correlation` 1.000, `peak_ratio_db` 0.00 dB); GAP-2 (Ardour VST3 plugin-parameter automation delivery) still-GAP with a sharper diagnosis — replacing Surge XT with the ACE Reverb LV2 exposed the same automation-non-delivery mechanism on LV2 as on VST3, so the gap is broader than the cycle-1 VST3-scoped diagnosis. Cycle 15 was a researcher pass framing the next fork. At cycle-15 exit the workspace holds 228 ledger events under the hardened schema, zero `promise_check` ERRORS, and 587 cross-branch integration test PASS lines including new §24 concat-hardening invariants; one pre-existing FAIL (M-RULES-1/extraction: provenance 28/76 resolvable) traces to the cycle-12 breadth expansion's provenance-resolution surface and is queued as a repair.

Introduction

By the end of cycle 12 the campaign had an end-to-end deterministic generation spine (M-GEN-1/first-generation on the cycle-9 28-row rules ledger), a hardened ledger-write emit sur-

face (cycle 10), a matching hardened `promise_check` surface (cycle 11), and a pipeline that had been demonstrated on multiple seeds. Three things were still open. The mechanical unlock the cycle-11 audit had named — grow the rules ledger by rerunning the extractors on the breadth seeds already on disk — was queued but not done; the last drift surface on the ledger-write side (`concat`) still bytes-appended shadow rows without re-validating them, producing recurring post-merge repair debt; and the two cycle-3 M-DAW-SPIKE-1 documented gaps (Ardour Lua MIDI-file import; Ardour VST3 plugin-parameter automation delivery) had aspirational fallback plans that had never been exercised end-to-end.

Cycle 13 was the researcher pass framing the three-clone fanout that would address all three. Cycle 14 was the worker pass that ran the three clones' underlying investigations. Cycle 15 was a follow-up researcher pass shaping the next fork.

Approach

Fork `ed041ef4c1dc` (three clones, disjoint file trees).

- **Clone 0 (M-RULES-1/breadth-expansion).** `scripts/rules/extract/breadth_seeds.py` walks the two breadth seeds' merged MusicXMLs and dispatches the five cycle-9 per-rule-type extractors under a `_common.py` context that names the seed and defaults to `synth_030s` behaviour when unset — so the extractor logic itself is byte-identical to cycle 9 and every cycle-9 anchor row reproduces without modification. A new `NullWithReason` helper returns `{"rule_type", "reason", "detail"}` rather than a `write_rule` call when content-incompatibility fires, so `scripts/rules/schema/*` stays frozen. `rule_id` continues as SHA-256 over canonical-JSON of `{rule_type, scope, sorted provenance_pointers, parameters}`, so cross-seed `rule_id` uniqueness is structurally guaranteed (different `provenance_pointers` → different hash). A separate `_salt_collision_analysis.py` runs the M-GEN-1 sampler at salts 0.4 against both the pre-expansion 28-row ledger and the post-expansion 76-row ledger and writes `data/rules/salt_collision_before_after.tsv` — out-of-band, no ledger mutation.
- **Clone 1 (`_infra/fanout-concat-hardening`).** The `concat` seam `long_exposure.workspace_bootstrap.concat_clone_ledgers(workspace: Path, fork_dir: Path) -> int` (the brief's `long_exposure.tools.fanout._concat_clone_ledgers` was a renamed reference; the worker documented the discrepancy rather than fabricating a module) was tightened to (a) pass every candidate row through `_ledger_schema.validate_event` before write, raising `LedgerConcatError` (subclass of `LedgerSchemaError`, subclass of `ValueError`) with field-named messages on drift, and (b) enforce per-milestone `ts` monotonicity within the candidate stream with a `content_hash_tiebreak` on exact `ts` collision — not file line number, which was the specific mechanism cycle 11's bug. The write is transactional via `NamedTemporaryFile + fsync + os.replace`. Public API surface is byte-preserved across cycles 1-11. A new `tests/test_fanout_concat_validation.py` (10 named cases) carries the mandatory `_LE_PARENT` `sys.path` shim so it runs cleanly under all three documented `PYTHONPATH` invocation flavors.
- **Clone 2 (M-DAW-SPIKE-1/gap-closure).** Cycle-3's coverage matrix (6 GREEN / 1 PARTIAL / 2 GAP over a five-axis × two-engine grid) documented aspirational fallback plans for the two GAPs but never exercised them. Clone 2 attempted each fallback end-to-end in the current environment (torch 2.13.0+cpu + torchvision 0.28.0 workaround already in place from earlier cycles).

Cycle 15 (researcher). Framed the next fork's shape from the cycle-14 outcomes and queued the surfaced repairs. No new load-bearing engineering.

Findings

M-RULES-1/breadth-expansion (clone 0) — validated/high

Ledger growth:

Seed	harmonic	rhythmic	melodic	form	arrangement	total	null-with-reason
seed_mid_50s	2	6	6	5	5	24	4 × harmonic / insufficient-progression (measure scope, unique_chords=1)
synth_060s	2	6	6	5	5	24	4 × harmonic / insufficient-progression (measure scope, unique_chords=1)
Total appended	4	12	12	10	10	48	8

Cycle-9 base 28 rows → post-expansion **76 rows** (target ≥ 15 rows exceeded 3×). Null-with-reason surfacing is honest, not evasive: on both breadth seeds the measure-scope harmonic extractor finds a single unique chord per window (unique_chords=1) and correctly reports insufficient-progression rather than fabricating a progression; song-scope harmonic still fires.

Regression contract preserved. head -28 data/rules/ledger.jsonl | sha256sum = 4fe722adde034c099ff9e65437f0d5c138cb3dd2595089960150af5c2546fc4b — matches cycle 9 bit-for-bit. Post-expansion whole-ledger SHA a6fd53e9bf9a10f6885888b0bb7d11a9a2aa97007e38ef0e6d47f4ef7d2857ae; two independent extraction runs produce byte-identical ledgers.

Salt-collision reduction. M-GEN-1 sampler at salts 0..4:

Colliding pair	Pre (28 rows)	Post (76 rows)
(0, 1) harmonic	☐	☐
(0, 1) rhythmic	☐	— resolved
(1, 2) melodic	☐	— resolved
(1, 4) arrangement (cycle-11-flagged)	☐	— resolved
(2, 3) rhythmic	☐	— resolved

Colliding pair	Pre (28 rows)	Post (76 rows)
(1, 4) rhythmic	—	□ (new)
(2, 4) melodic	—	□ (new)
(3, 4) arrangement	—	□ (new)

The specific cycle-11-flagged (salt 1, salt 4) arrangement collision resolves (arrangement rule_id shifts from rule_b75cc391f671037a under both salts pre-expansion to rule_b99a5066e653b247 at salt 1 and rule_a8ffe2f88dc29eed at salt 4 post-expansion). Overall pair count drops 5 → 4 (~20 %) — sub-proportional to the 3× pool growth. Three of the four post-expansion pairs involve salt = 4, strongly suggesting the residual rate reflects something specific about salt = 4’s hash-space geometry on this rule space rather than a general pool-size effect. The report surfaces “structural diversity within a rule_type may matter more than raw pool size” as a research signal for the next cycle rather than tuning the collision count down.

Auditor decision: **VALIDATED / COMPLETE** at /high — mechanical target hit and cycle-9 regression contract preserved; the sub-proportional overall reduction is honestly documented rather than concealed.

_infra/fanout-concat-hardening (clone 1) — validated/high

All ten sufficiency criteria pass under the auditor’s live re-verification:

Criterion	Evidence
Concat seam located and modified in place, public API unchanged LedgerConcatError(LedgerSchemaError) real subclass, MRO verified	inspect.signature(concat_clone_ledgers) → (workspace: Path, fork_dir: Path) -> int issubclass(LedgerConcatError, LedgerSchemaError) and issubclass(LedgerConcatError, ValueError) both True
All existing ledger rows pass tightened concat with no schema grandfathering	Case 8 + live re-verification 228 rows
Two documented drift patterns rejected with field-named messages	Case 2 (missing event_id, cycle-10 pattern), Case 5 (per-milestone ts monotonicity, cycle-11 pattern)
Concat byte-deterministic and idempotent	Cases 6 + 7 + live dogfood on a real shadow ledger (4 rows first, 0 rows second)
All existing worker-side test suites remain green New test file runnable in 3 PYTHONPATH flavors via _LE_PARENT shim SSoT is-identity	Writer 13/13; integration 587 PASS (1 pre-existing unrelated FAIL) All three flavors 10/10 promise_check.REQUIRED_EVENT_FIELDS is _ledger_schema.REQUIRED_EVENT_FIELDS
docs/fanout_concat_hardening.md with all 7 required sections	Grep confirms
_infra/fanout-concat-hardening in plan_of_record.md 5-col Milestones table	Ledger events resolve; no plan-file drift ERROR

The one honest carve-out (\$5 of the docs). Applying per-milestone-file-order ts monotonicity retroactively to the main ledger as a candidate stream surfaces 7 pre-existing cycle-1-era

violations plus 11 ts collisions. Resolution: invariant scope is candidate-stream only — main ledger rows are grandfathered against monotonicity but not against schema (all 222 rows pass schema). This matches real tool usage — no fan-out re-ingests the main ledger as a candidate — avoids the fabricate-repair-ts trap cycle 11 fell into, and preserves the brief’s “no schema grandfathering” rule.

M-DAW-SPIKE-1/gap-closure (clone 2) — validated/medium; parent M-DAW-SPIKE-1 remains validated/high

Cycle-3 baseline 6 GREEN / 1 PARTIAL / 2 GAP → cycle-14 8 GREEN / 1 PARTIAL / 0 GAP / 1 redefined-GAP (five-axis × two-engine matrix).

GAP	Fallback exercised	Verdict	Evidence
GAP-1 (Ardour Lua MIDI-file import)	Fallback #2 — fluidsynth pre-render + hand-authored Source/Region/Playlist audio-region XML	redefined-GAP	env_correlation = 1.000, peak_ratio_db = 0.00 dB (data/daw_spike/gap1_ midi_import_ measurement.json)
GAP-2 (Ardour VST3 plugin-parameter automation delivery)	Fallback #2 — replace Surge XT Effects VST3 reverb with ACE Reverb LV2, author wet-mix automation	still-GAP with sharper diagnosis	second/first RMS ratio = 1.0000 vs cycle-1 baseline 2.05 / DawDreamer reference 2.46 (data/daw_spike/gap2_ lv2_measurement.json); Ardour Lua plugin_automation() fails to deliver on LV2 as well as VST3 — gap is broader than the cycle-1 VST3-scoped diagnosis

GAP-1’s redefined-GAP is the honest kind of closure: the primary Ardour Lua MIDI-file import path was in fact reachable via a different mechanism than cycle 3 had documented (pre-render + audio-region), and the numerical fidelity of that mechanism to the target rendering is perfect. GAP-2’s still-GAP is the honest kind of non-closure: the fallback was exercised in full, the same automation-non-delivery mechanism fired on LV2 as on VST3, and the report’s contribution is the sharpened diagnosis — the gap is not VST3-specific — rather than a claim of closure.

Parent M-DAW-SPIKE-1 stays validated/high per cycle 3; this cycle’s contribution is axis-level detail.

Cycle-15 researcher pass

Framed the next fork’s shape from the three clones’ outputs. Recorded the pre-existing FAIL (M-RULES-1/extraction: provenance 28/76 resolvable) — the ratio traces to how the cycle-12 breadth expansion’s provenance_pointers resolve against the cycle-9 anchor prefix and is

a repair, not a defect uncovered here — as queued for a future cheap cycle. Queued the cycle-11-audit-named probes on the salt = 4 residual (salts 5..9 on the 76-row ledger) and the structural-diversity axis (a non-F_major seed) as candidate cycle-16 work.

Campaign-level state at cycle-15 exit

- **Ledger:** 228 events under the SSoT schema. Emit, check, and concat surfaces now all consume the same `_ledger_schema` module with is-identity of `REQUIRED_EVENT_FIELDS` verified live.
- **Rules ledger:** 76 rows across three seeds, cycle-9 anchor prefix preserved byte-for-byte.
- **Tests:** `test_ledger_writer_validation.py` 13/13; `test_fanout_concat_validation.py` 10/10 × 3 PYTHONPATH flavors; `test_integration_cross_branch.py` 587 PASS with \$24 concat-hardening invariants; one pre-existing unrelated FAIL on M-RULES-1/extraction: provenance 28/76 resolvable queued.
- **Validators:** `promise_check` 0 ERRORS, only pre-existing WARNs (the shadow-ledger orphan-artifact WARN on `tests/test_fanout_concat_validation.py` clears at the next fork-conductor merge under the dogfood of clone 1's own tightened concat). `org_check` no new WARNs.
- **Blocked on rated audio:** parent M-EAR-1 v0 training; M-INGEST-1/egress-ready-automation is IDLE and awaits its two-consecutive-media_ok=true trigger.

Discussion

Three things about this range are worth naming.

First, the ledger-write triangle is now complete. Cycle 10 hardened the emit surface, cycle 11 hardened the check surface, and this range's clone 1 hardened the concat surface. All three now route through the same `SSoT_ledger_schema` module, and the last remaining bypass — the `_repair_and_emit_*` direct-append pattern from cycle 10 — is queued for retirement rather than tolerated as a permanent exception. The dogfood confirmation that clone 1's own shadow-ledger closure event will land through the very concat seam it hardened is a satisfying end-to-end proof of the migration claim, and it means the recurring post-merge integration debt from cycles 10 and 11 (roughly one worker-cycle each on shadow-ledger surgery) should now approach zero on ledger-drift shapes. The next fork's post-merge integration will be the ground-truth test of that reduction.

Second, the M-RULES-1 breadth expansion is the campaign's cleanest example so far of a mechanical unlock that hits its named target and then honestly surfaces a residual finding. The cycle-11 audit named the (salt 1, salt 4) arrangement collision as the unlock; this range resolved that specific collision, and the report did not claim more than that. The overall salt-collision pair count dropped only from 5 to 4 despite the pool tripling, and three of the four post-expansion pairs involve salt = 4 — which strongly suggests something specific about salt = 4's hash-space geometry rather than a general pool-size effect. The report surfaces “structural diversity within a rule_type may matter more than raw pool size” as a hypothesis to test in a future cycle rather than tuning the collision count down. This is the falsifiability discipline paying off in the exact case it was designed for: the mechanical target hits, the mechanism the audit named is closed, and the residual is not forced into an artificially clean number.

Third, clone 2's split verdict on the two M-DAW-SPIKE-1 gaps is worth preserving as a canonical example of the three-way outcome distinction the campaign relies on: GREEN (the primary path works within a documented tolerance to a target), still-GAP (the fallback was exercised end-to-end and did not achieve the target, with a specific diagnosis of why), and redefined-

GAP (the primary path was actually reachable via a different mechanism than originally documented, and the numeric fidelity of that mechanism is measured against the target). GAP-1's redefined-GAP produces perfect fidelity (`env_correlation = 1.000`, `peak_ratio_db = 0.00 dB`); GAP-2's still-GAP produces a sharper diagnosis (the automation-non-delivery is broader than VST3-scoped) rather than a claim of closure. Two very different kinds of research outcome, both honestly delivered.

The uncalibrated CORN head remains the campaign's biggest open credibility gap. The `ear.calibration = "synthetic_labels_only"` sentinel prevents any M-GEN-1 pass's rating from being read as a musical judgment, and the M-INGEST-1/egress-ready-automation state machine will fire the retraining pipeline unattended the moment two consecutive fresh `media_ok=true` rows land. Nothing in this range changes that; nothing in this range needed to.

Open Questions

- **Cycle-13 batch-v2 rerun on the 76-row ledger.** The live salt = 0 selection will change for melodic / form / arrangement on the expanded ledger; cycle-11 batch-v1 anchors remain pinned in a saved `sampling_manifest.json` and \$23 of the cross-branch integration test still passes reading that JSON. The next cycle must expect and document this — it is not a bug.
- **Salt = 4 over-representation.** Three of four post-expansion collision pairs involve salt = 4. Probe with salts 5..9 on the 76-row ledger to distinguish hash-space geometry for small-N pools from a salt-4-specific effect. Roughly 250 sample cells to characterise (5 `rule_types` × 5 additional salts × ~10 candidates each).
- **Structural-diversity bottleneck hypothesis.** The 3× pool growth produced only a ~20 % collision-rate reduction; a non-F_{major} seed with different instrumentation would test the structural-diversity mechanism cheaply.
- **M-RULES-1/extraction: provenance 28/76 resolvable** repair. Pre-existing FAIL surfaced by clone 1's audit; traces to the cycle-12 breadth expansion's provenance-resolution surface. Small cheap repair; queued.
- **Retire _repair_and_emit * direct-append callers** so `concat`, `emit`, and `check` are the only three ways a row enters the ledger.
- **Tighten manually-set event_id against content-hash mismatch** — a row whose `event_id` does not derive from its own content is currently accepted at `concat`; catching that is the next legibility win.
- **Multi-fork parallel concat race.** Atomic `os.replace` protects the write, but two concurrent `concat` calls on the same workspace with disjoint fork directories have no cross-lock; low priority, no known live occurrence.
- **Pre-flight brief-linter** that resolves every named seam in a research brief against the actual module tree (recurring cycles-10/11/12 lesson: clone 1's brief pointed at `long_exposure.tools.fanout._concat_clone_ledgers`, actual seam was `long_exposure.workspace_bootstrap.concat_clone_ledgers`).
- **GAP-2 fallback #1** on M-DAW-SPIKE-1 — read `libs/ardour/plugin_insert.cc` for the missing Lua-side automation-arming call. Left open by clone 2.
- **ardour_region_xml.py extraction** — the audio-region XML fragment in `gap_closure_midi_import.py` is stable enough to promote to `scripts/daw_spike/ardour_region_xml.py` when a second call-site needs it.
- **DawDreamer plugin catalog expansion** — with the torchvision workaround live, breadth-probing Dragonfly / MVerb / LSP LV2 reverbs could seed an M-GEN-1/batch-v2+ effects chain disjoint from the cycle-9 pinned chain.
- **CORN-head calibration** — still blocked on rated audio; will fire unattended through M-INGEST-1/egress-ready-automation.

Appendix: Provenance

Cycle range: cycles 13-15. **Working directory:** /home/user/long-exposure-runs/music-gen.
Session references: cycle 13 researcher a235a90e-2f14-4196-999d-2da2848f36b3; cycle 14 worker 635e473a-32ac-4f27-9497-6ad6d680251f; cycle 15 researcher 036d68ca-b9cd-4d43-af0e-0b81620fb08c.

Sub-agent transcripts (fork ed041ef4c1dc clones).

- Clone 0 (M-RULES-1/breadth-expansion): researcher fda6bcd-fbb1-46be-bd7c-f2c98d2f43c1, worker 850df679-dc12-4fa2-929a-e77aabe88691, auditor f58047a6-cd08-491a-9388-e3da70488aca. Auditor decision COMPLETE; sub-milestone closes at validated/high.
- Clone 1 (_infra/fanout-concat-hardening): researcher b1a99d47-7d78-4c5c-b3df-4532291a64fc, worker 2703c070-3f84-4339-928e-596684aef14a, auditor 4bba1416-da3b-4976-9e82-7ff95e786e36. Auditor decision VALIDATED; sub-milestone closes at validated/high.
- Clone 2 (M-DAW-SPIKE-1/gap-closure): sub-milestone closes at validated/medium; parent M-DAW-SPIKE-1 remains validated/high.

Deliverables on disk at cycle-15 exit.

- Clone 0: scripts/rules/extract/breadth_seeds.py + _common.py extended (schema untouched); data/rules/ledger.jsonl 28 → 76 rows; data/rules/breadth_expansion_summary.json; data/rules/salt_collision_before_after.tsv; docs/figures/rules_extraction_breadth_growth.png; docs/rules_extraction_breadth_report.md (315 lines).
- Clone 1: seam tightening in long_exposure/workspace_bootstrap.py; LedgerConcatError and content_hash_tiebreak added to long_exposure/tools/_ledger_schema.py; tests/test_fanout_concat_validation.py (341 lines, 10 cases, _LE_PARENT shim); §24 of the cross-branch integration test; docs/fanout_concat_hardening.md (185 lines, 7 sections).
- Clone 2: scripts/daw_spike/ (new directory, 6 files); data/daw_spike/{coverage_matrix_v2.json, gap1_midi_import_measurement.json, gap2_lv2_measurement.json, gap_closure_lv2_render.wav, gap_closure_lv2_state.json, gap_closure_midi_prerender.wav, gap_closure_midi_render.wav, sessions/gap_closure_{lv2,midi}/...}; docs/daw_spike_gap_closure_report.md; docs/figures/daw_spike_coverage_v2.png.

Load-bearing SHAs at cycle-15 exit.

```
head -28 data/rules/ledger.jsonl | sha256sum
→ 4fe722adde034c099ff9e65437f0d5c138cb3dd2595089960150af5c2546fc4b (cycle-9 anchor, preserved)

sha256 data/rules/ledger.jsonl
→ a6fd53e9bf9a10f6885888b0bb7d11a9a2aa97007e38ef0e6d47f4ef7d2857ae (post-expansion 76 rows)
```

Ledger routing. Clone 0 emitted 8 shadow events (plan-registration, kickoff, per-seed × 2, parent closure, integration-test extension, archive, _run/clone-0-scope-complete); clone 1 emitted 6 shadow events; clone 2 emitted 5 shadow events. Post-integration, promise_check reports 228 rows total, 0 ERRORS, 26 pre-existing WARNs unchanged, with one MINOR orphan-artifact WARN on tests/test_fanout_concat_validation.py that clears automatically when the fork conductor collapses the shadow ledger via clone 1's own tightened concat (dogfood proof).

Public API preserved. inspect.signature(concat_clone_ledgers) → (workspace: Path, fork_dir: Path) -> int byte-for-byte with cycles 1-11.

SSoT identity verified live. promise_check.REQUIRED_EVENT_FIELDS is _ledger_schema.REQUIRED_EVENT_FIELDS → True (emit + check + concat now consume the same module).

Environment stack unchanged since cycle 10. mscore3 3.2.3 headless; Python 3.11.15;

numpy 1.26.4; music21 9.1.0; mir_eval 0.8.2; fluidsynth (Debian) with pinned SF2 74594e8f...1cb0; DawDreamer + Surge XT Effects.vst3 at /usr/lib/vst3/; basic-pitch 0.4.0 in workspace/basic_pitch_venv/; Ardour 8.x for the M-DAW-SPIKE-1 GAP-closure attempts (with the recorded caveat that session-cleanup SIGABRT is intermittent but the render WAV bytes are always committed before cleanup, so post-merge validators should not gate on ardour8-export return-code alone). Single-thread BLAS pins throughout.

Rated audio. Still egress-blocked per corpus/CORPUS_STATUS.md. M-INGEST-1/egress-ready-automation state machine remains IDLE; runtime state files correctly absent until the first live trigger. Not this range's problem; the machine is pre-wired.

Handoff to next cycle. The queued natural next steps are the salt = 4 probe (salts 5..9 on the 76-row ledger), the structural-diversity axis test (a non-F_major seed), the M-RULES-1/extraction: provenance 28/76 resolvable repair, and the _repair_and_emit_* retirement. Anything requiring rated audio is a straight-line consequence of the egress-ready state machine firing.